MapLibre ArcGIS Plug-in: Building web apps

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TECHNOLOGY SUMMIT

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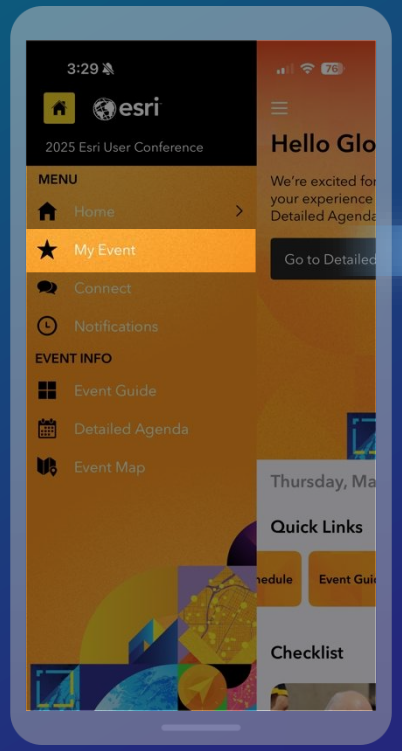

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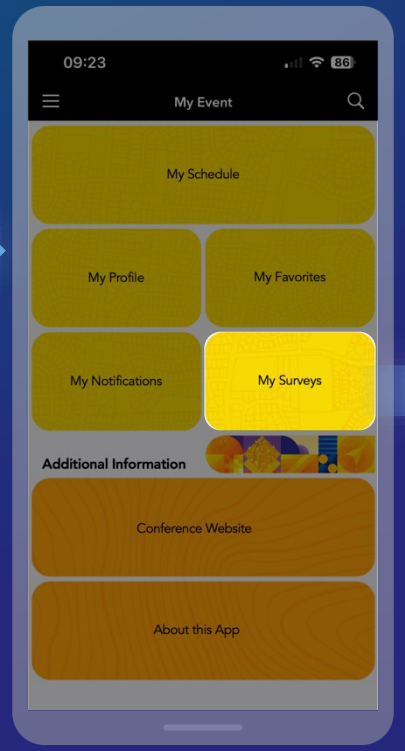
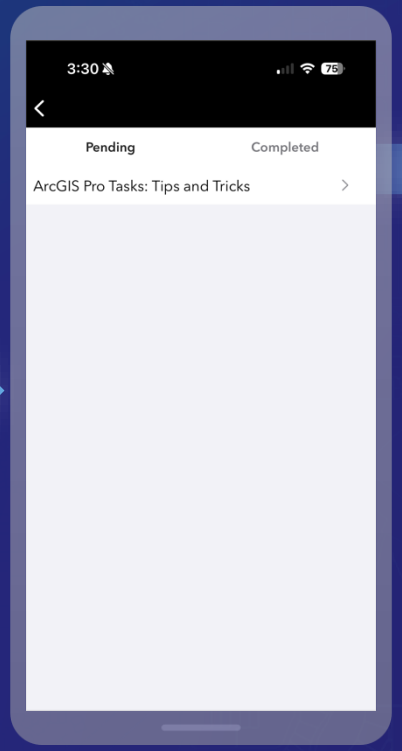
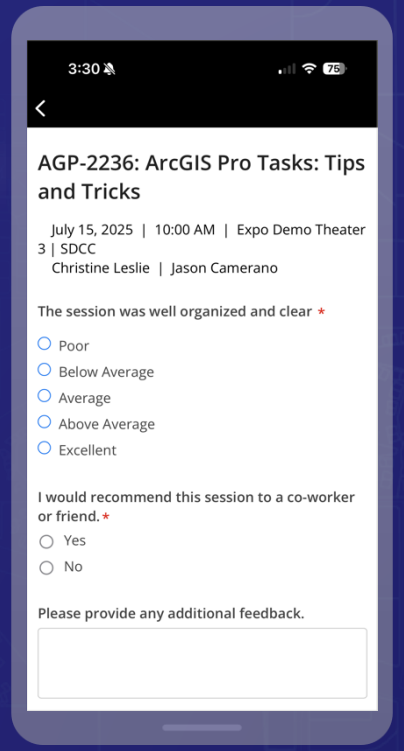


Select the desired survey



Complete the survey



Agenda

- MapLibre GL JS
- Introducing MapLibre ArcGIS
- Q&A

MapLibre GL JS

MapLibre GL JS

- Javascript mapping library for web browsers
- Fully open source under BSD 3-Clause license
- Created in 2020, fork of Mapbox GL JS 1.13
- 10k stars on GitHub, 1.8 mil weekly NPM downloads
- Supported by Esri, Amazon

MapLibre

MapLibre GL JS

MapLibre GL JS: Technical features

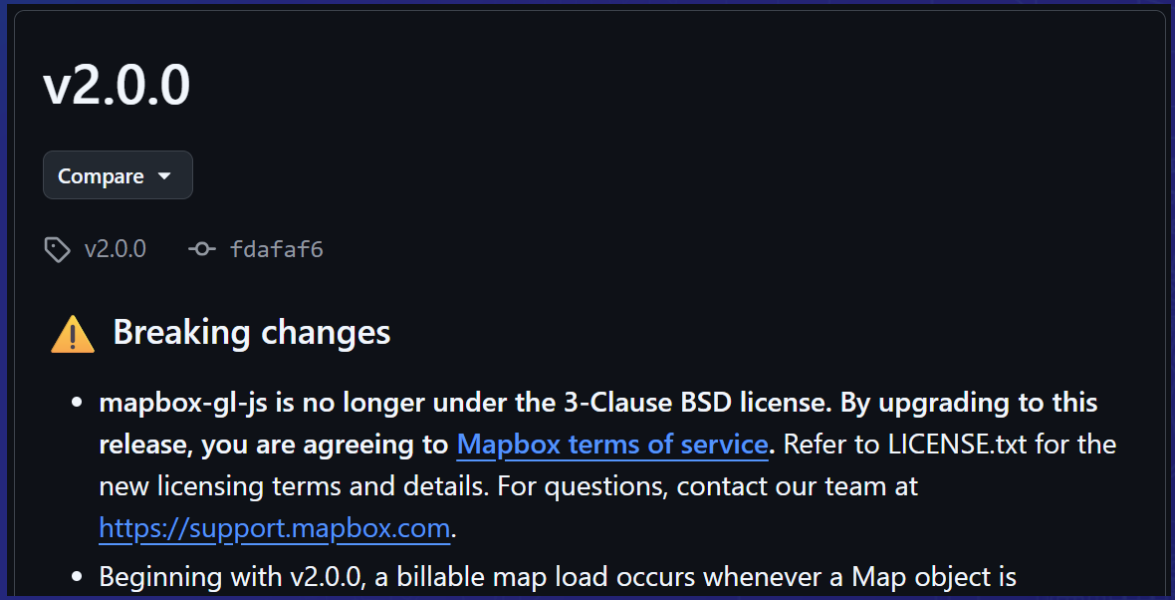
- True open source (BSD-3-Clause)
- High performance, low footprint
- WebGL-native infrastructure
- Built for rendering vector data, scalable
- Easy, data-driven styling

MapLibre

MapLibre GL JS

History lesson: mapbox-gl-js

- Originally created by Mapbox
- 1.0 versions are fully open-source
- 2.0 release: Re-licensed to proprietary software



v2.0.0

Compare

v2.0.0 fdafaf6

⚠️ Breaking changes

- **mapbox-gl-js is no longer under the 3-Clause BSD license. By upgrading to this release, you are agreeing to Mapbox terms of service.** Refer to LICENSE.txt for the new licensing terms and details. For questions, contact our team at https://support.mapbox.com.
- Beginning with v2.0.0, a billable map load occurs whenever a Map object is

- Community response: MapLibre GL JS

Style Specification

- Originally created by Mapbox
- Also forked by MapLibre, currently v8
- Fully supported by ArcGIS
- ArcGIS Vector Tile Style Editor

```
{
  "version": 8,
  "sources": {
    "esri": {
      "type": "vector",
      "url": "https://basemaps.arcgis.com/arcgis/rest/services/World_Basemap_v2
      "tiles": [
        "https://basemaps.arcgis.com/arcgis/rest/services/World_Basemap_v2/Vect
      ]
    }
  },
  "layers": [
    {
      "id": "background",
      "type": "background",
      "layout": {

      },
      "paint": {
        "background-color": "#a7d6fe"
      }
    },
    {
      "id": "Land/Not ice",
      "type": "fill",
      "source": "esri",
      "source-layer": "Land",
```

MapLibre Style Specification

Introducing: MapLibre ArcGIS

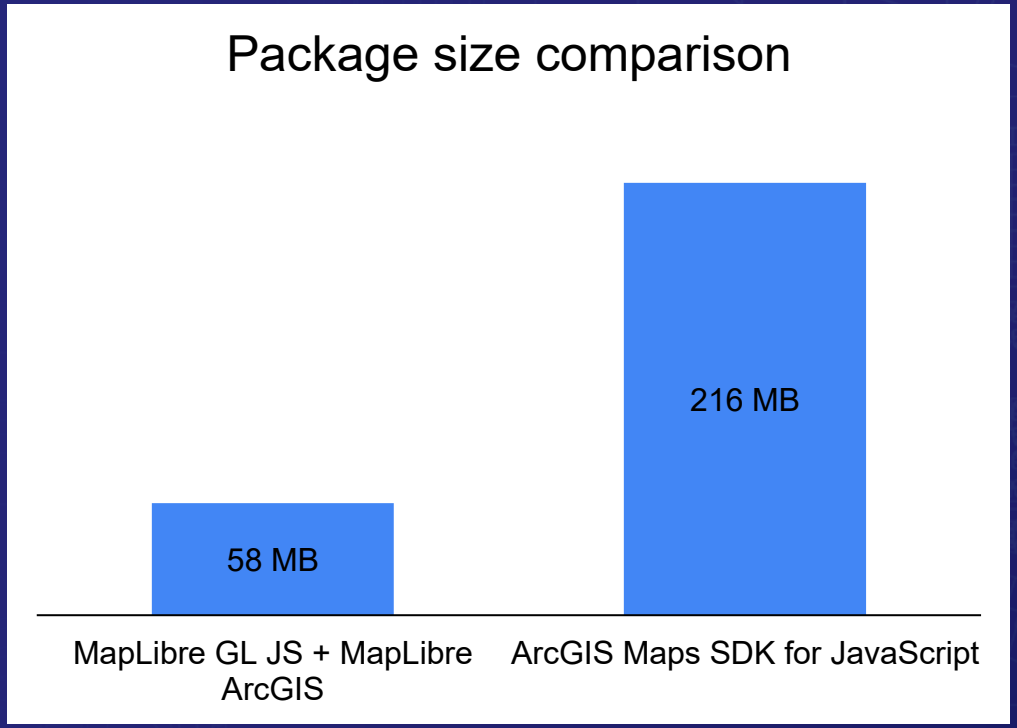
MapLibre ArcGIS

- Official ArcGIS integration with MapLibre GL JS
- Standalone plugin
- Key features:
  - ArcGIS basemaps
  - Feature layers
  - Vector tile layers
- Handles data attribution

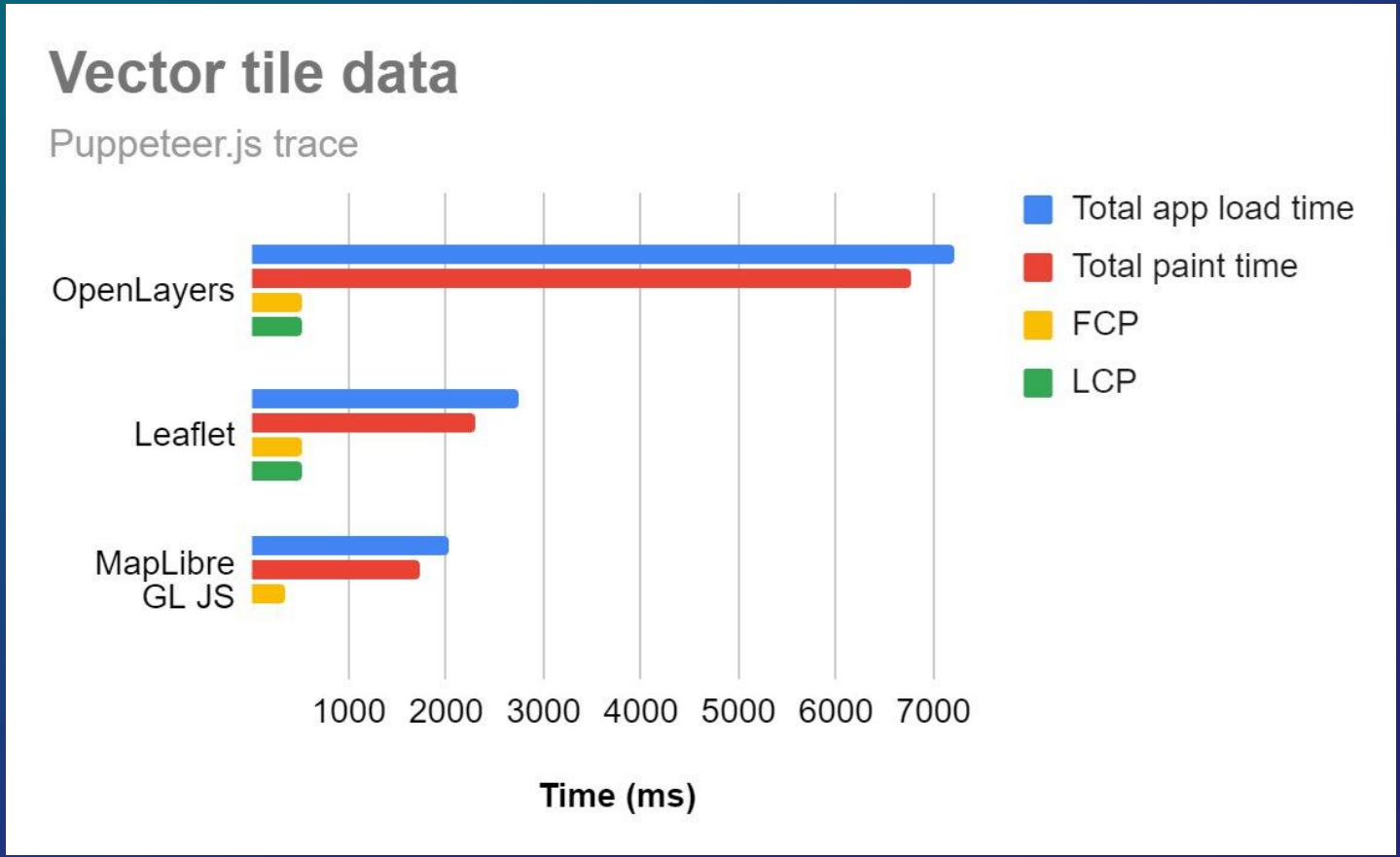
MapLibre ArcGIS: Package size

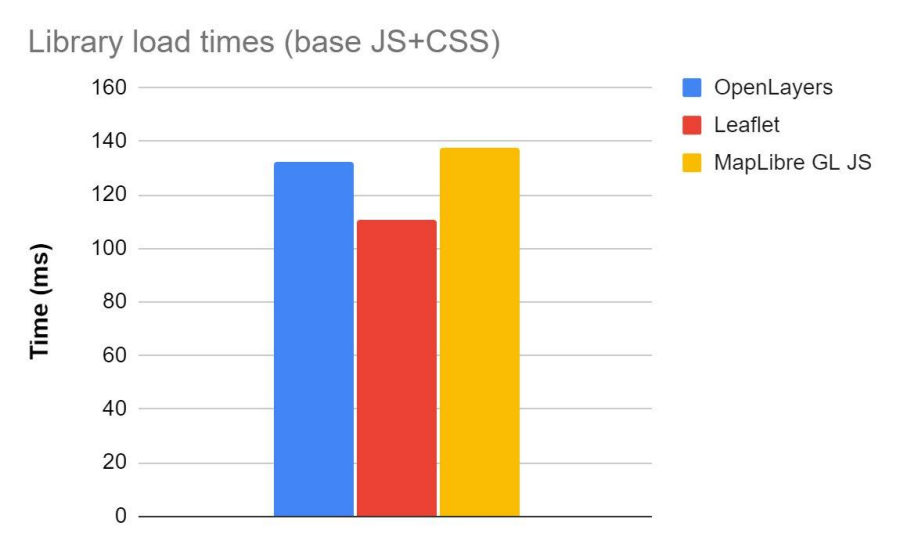
- MapLibre ArcGIS (Minified + gZipped): 258.8 kB
- MapLibre GL JS: (installed w/ dependencies): 56 MB

- Compare with ArcGIS JS Maps SDK:
  - 216 MB



MapLibre ArcGIS: Performance

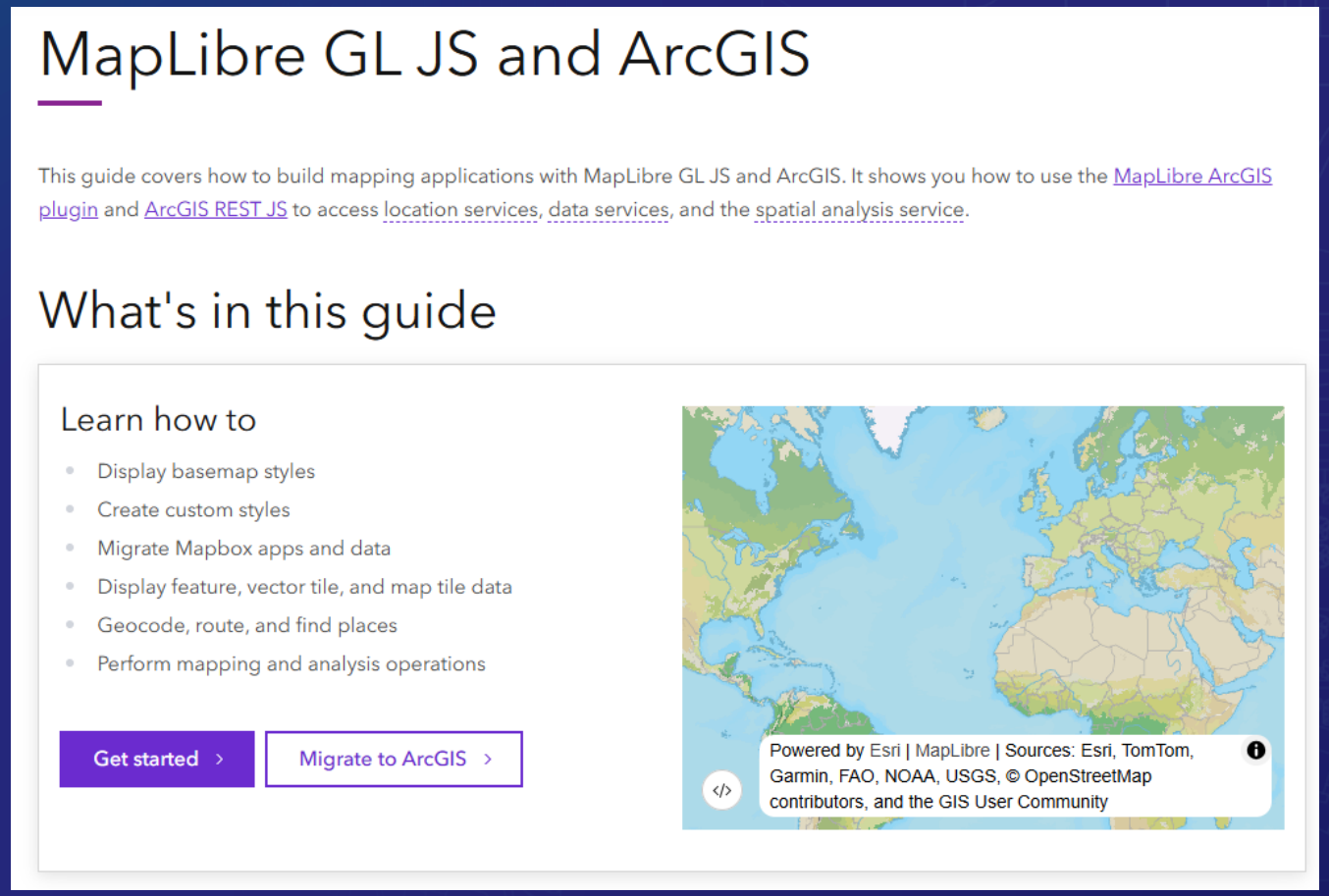




- * performance data recorded in 2023
- https://github.com/ak-kemp/foss4g-comparison

Developer guide

- Complete documentation: https://developers.arcgis.com/maplibre-gl-js/
- API reference
- 30+ tutorials
- Migration guide: Mapbox to ArcGIS



MapLibre GL JS and ArcGIS

This guide covers how to build mapping applications with MapLibre GL JS and ArcGIS. It shows you how to use the MapLibre ArcGIS plugin and ArcGIS REST JS to access location services, data services, and the spatial analysis service.

What's in this guide

Learn how to

- Display basemap styles
- Create custom styles
- Migrate Mapbox apps and data
- Display feature, vector tile, and map tile data
- Geocode, route, and find places
- Perform mapping and analysis operations

Get started › Migrate to ArcGIS ›

Powered by Esri | MapLibre | Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

Capabilities: Basemaps

- Load basemaps from ArcGIS Basemap Styles Service
- 50+ styles available
- Vector and raster tiles
- Served as MapLibre style JSON

```
const map = new maplibregl.Map({
  container: "map", // the id of the div element
  zoom: 5, // starting zoom
  center: [138.2529, 36.2048], // starting location
});

const basemapStyle = maplibreArcGIS.BasemapStyle.applyStyle(map, {
  style: 'arcgis/imagery',
  token: apiKey
});
```

Capabilities: Basemap Sessions

- MapLibre ArcGIS supports authentication with basemap sessions
- Extremely cost-effective way to load basemaps
- Requires an ArcGIS Location Platform account
- Auto-refresh functionality

```
const basemapSession = await maplibreArcGIS.BasemapSession.start({
  styleFamily: 'arcgis',
  token: apiKey, // ArcGIS API key
  duration: 43200, // 12 hours in seconds
  autoRefresh: true
});
```

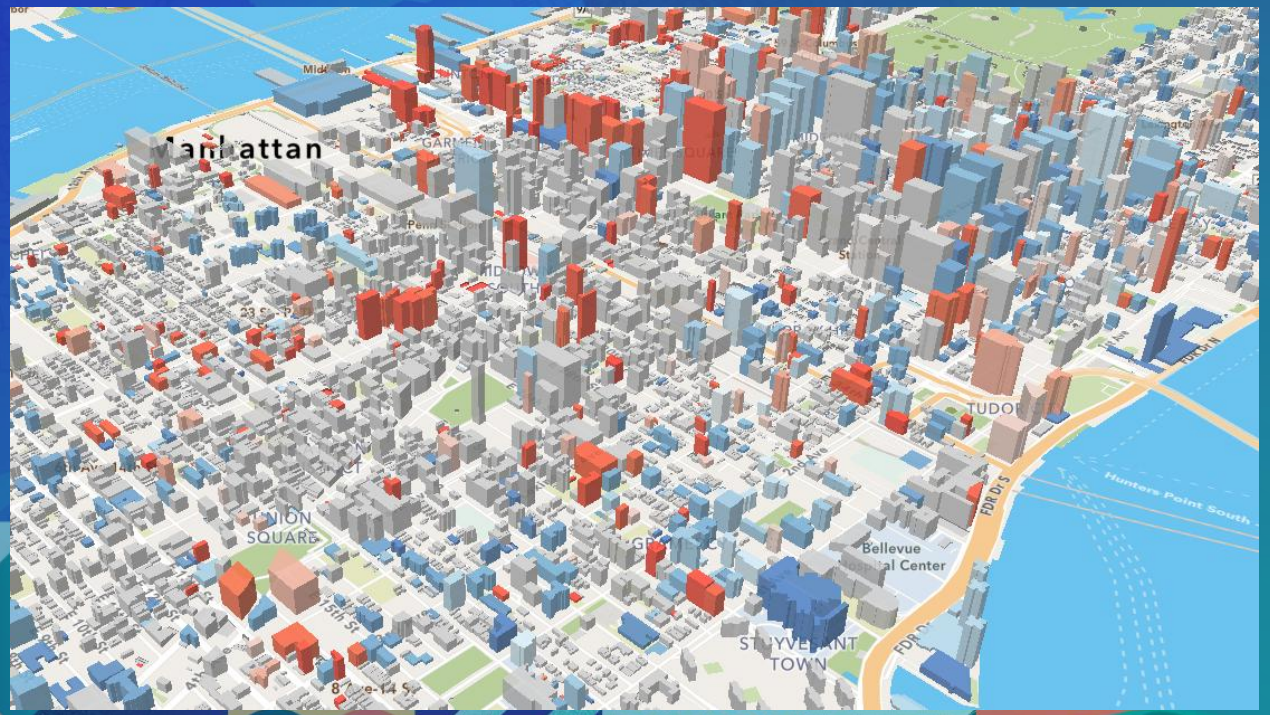
Demo: Basemaps w/ Basemap Session

MapLibre ArcGIS

Capabilities: Feature layers

- Load features from an item ID or service URL
- New update: all data transferred as PBF -> 5-10x more efficient than GeoJSON
- Handles extremely large layers
- Multi-layer support

```
// Load parks from from portal item ID
const parks = await maplibreArcGIS.FeatureLayer.fromPortalItem('f2ea5d874dad427294641d2d45097c0e');
parks.addSourcesAndLayersTo(map);
```



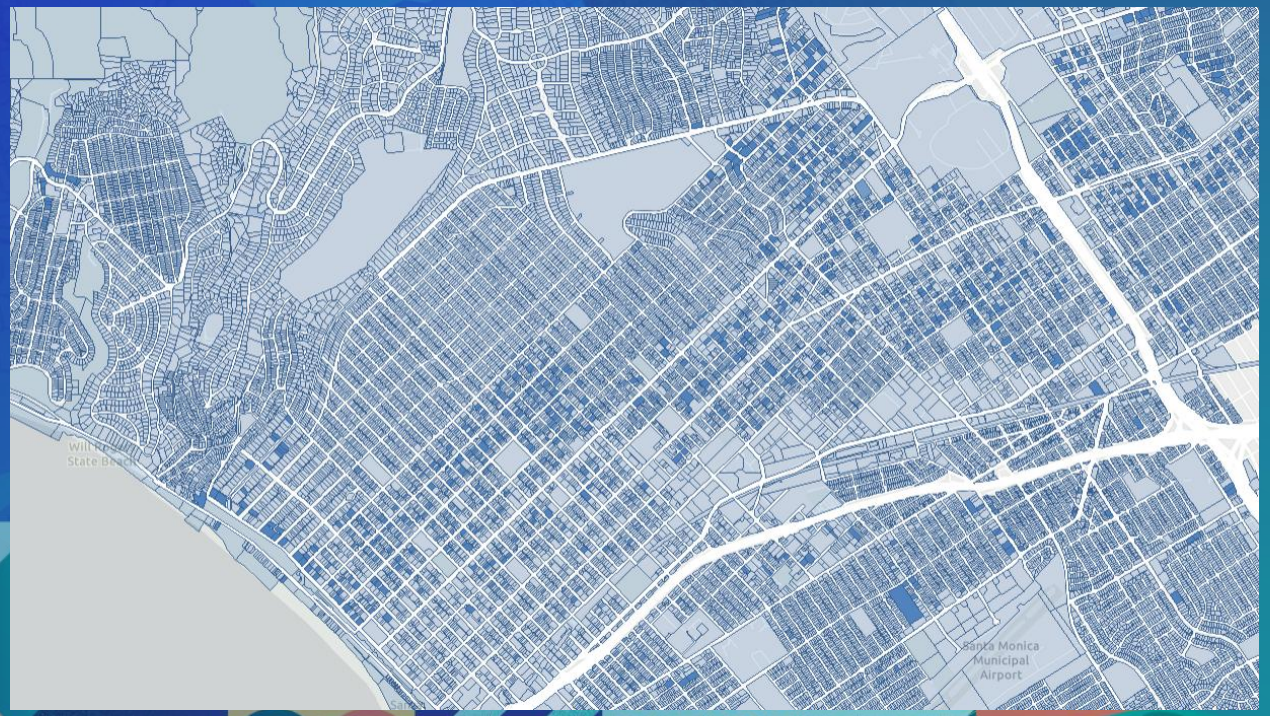
Demo: Feature layer

MapLibre ArcGIS

Capabilities: Vector tile layers

- Load vector tiles from an item ID or service URL
- Native support in MapLibre ecosystem
- Automatically load styles saved in item
- Edit styles using the ArcGIS Vector Tile Style Editor

```
// Add parcels vector data from Item ID
const vectorLayer = await maplibreArcGIS.VectorTileLayer.fromPortalItem('f0298e881b5b4743bbdf2c7d378acc84');
vectorLayer.addSourcesAndLayersTo(map);
```



Demo: Vector tiles

MapLibre ArcGIS

Other supported open-source libraries

- Esri Leaflet – a popular alternative
- OpenLayers – best for GIS
- CesiumJS – best for 3D

- MapLibre ArcGIS – best for basemaps, vector tiles, feature layers

Conclusion

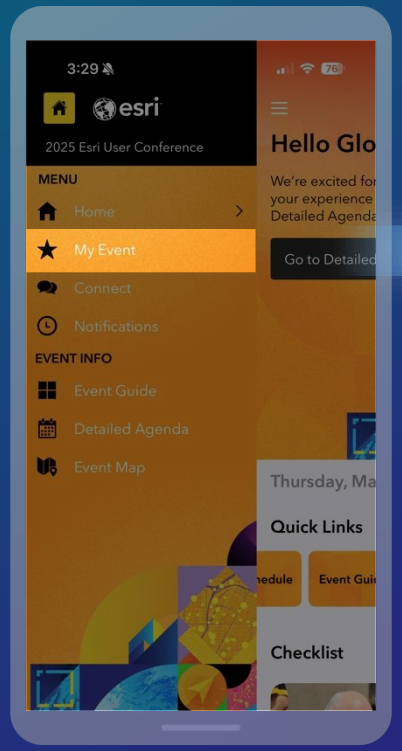
- MapLibre ArcGIS is a robust, performant alternative for web environments
- Tightly integrated with ArcGIS
- Perfect for visualizing data
- Active development, more features being added
- Maps SDK for JavaScript is still best-in-class for GIS analysis

Resources

- https://github.com/EsriDevEvents/maplibre-arcgis-building-web-apps-2026
- ^ Slides + resources
- ArcGIS Vector Tile Style Editor

Please share your feedback in the app

Select “My Event” in the event app main menu



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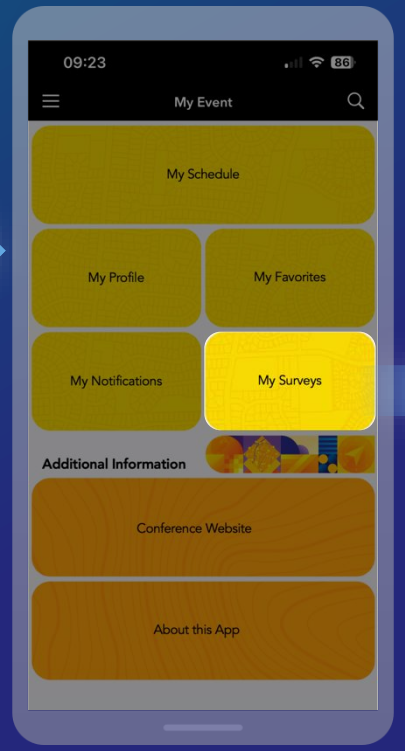
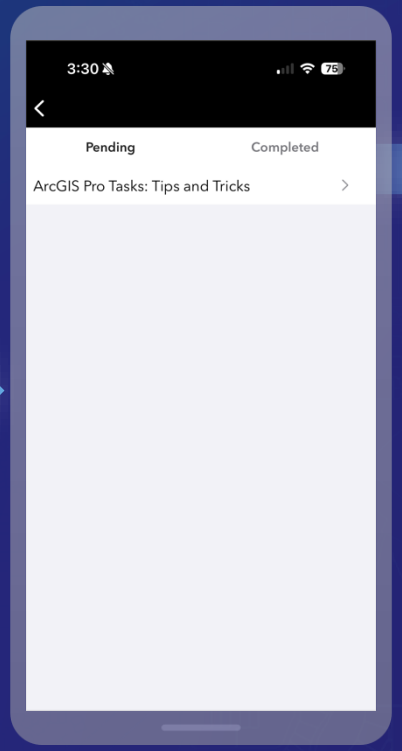
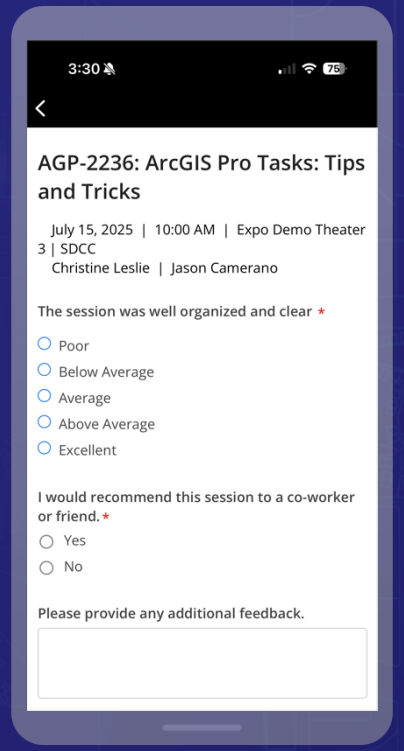


Select the desired survey



Complete the survey



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